Entity Embedding

ACTL3143 & ACTL5111 Deep Learning for Actuaries

Patrick Laub

Lecture Outline

- **Entity Embedding**
- The French Motor Dataset
- From One-Hot Encoding to Embeddings
- Keras' Functional API
- French Motor Dataset with Embeddings
- Scale by Exposure

Each category used to be a column

When we preprocessed categorical variables, *one-hot encoding* gave a *column for every category*, so you could always read off what each number meant:

Category	One-hot vector
`R11`	$[1, 0, 0, \ldots, 0]$
`R24`	$[0, 1, 0, \ldots, 0]$
`R93`	$[0, 0, \ldots, 1, 0]$

Each column *means one specific category*, but the vector is *sparse* (mostly zeros) and *long* (one entry per category), and every category sits the *same distance* from every other.

Entity embeddings: short, dense & learned

An *entity embedding* (Guo & Berkhahn, 2016) replaces that sparse column-per-category block with a *short, dense vector* that the network *learns during training*:

$$\text{R11} \longrightarrow [0.07,\ -0.42], \qquad \text{R24} \longrightarrow [0.11,\ -0.39]$$

- the vector is *short* — e.g. 2 numbers instead of 22 columns;
- the dimensions are *learned*, not categories — there is no "`R11` column";
- categories with a *similar effect on the target* tend to cluster together.

This is particularly useful when the categorical variable can take on a large number of different levels (it has high cardinality). Though, in that situation, *rare* levels are fit from very little data, so they may not be placed optimally.

It's the same idea as word embeddings

- A *word embedding* is just an entity embedding where the "entity" is a *word*.
- *Entity embedding* applies the same trick to *any* categorical variable: region, vehicle brand, occupation, postcode, ...
- Typically word embeddings are *pre-trained* on huge text corpora — i.e. it is a *transfer learning* technique — though here we have labels so learn the embeddings as part of the supervised task.

They are both, in essence, a *lookup table of learned vectors*.

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Imports needed for these demos

```
import random
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

from keras.models import Sequential, Model
from keras.layers import Dense, Input, Embedding, Reshape, Concatenate
from keras.callbacks import EarlyStopping
from keras.utils import plot_model

from sklearn import set_config
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder, StandardScaler
from sklearn.compose import make_column_transformer

set_config(transform_output="pandas")
```

Revisit the French motor dataset

	IDpol	ClaimNb	Exposure	Area	VehPower	VehAge	DrivAge	Bonu
0	1.0	1.0	0.10000	D	5.0	0.0	55.0	50.0
1	3.0	1.0	0.77000	D	5.0	0.0	55.0	50.0
...	...	...	...	...	...	...	...	...
678011	6114329.0	0.0	0.00274	B	4.0	0.0	60.0	50.0
678012	6114330.0	0.0	0.00274	B	7.0	6.0	29.0	54.0

678013 rows × 12 columns

Source: Noll et al. (2020).

Data dictionary

Variable	Description	Preprocessing
`IDpol`	Policy number (unique identifier)	Dropped
`ClaimNb`	Number of claims on the given policy	**Target**
`Exposure`*	Total exposure in yearly units	Normalised
`Area`	Area code (ordinal)	Ordinal Encode
`VehPower`	Power of the car (ordinal encoded)	Normalised
`VehAge`	Age of the car in years	Normalised
`DrivAge`	Age of the (most common) driver in years	Normalised
`BonusMalus`	Bonus–malus level between 50 and 230 (with reference level 100)	Normalised
`VehBrand`*	Car brand (nominal)	One-hot
`VehGas`	Diesel or regular fuel car (binary)	One-hot
`Density`	Density of inhabitants per km² in the city of the living place of the driver	Normalised
`Region`*	Regions in France (prior to 2016)	One-hot

** The three variables we single out later: `Region` & `VehBrand` get embeddings, and `Exposure` becomes an offset.*

Source: Noll et al. (2020).

The model

Have $\{(\mathbf{x}_i, y_i)\}_{i=1,\ldots,n}$ for $\mathbf{x}_i \in \mathbb{R}^{47}$ and $y_i \in \mathbb{N}_0$.

Assume the distribution

$$Y_i \sim \mathsf{Poisson}(\lambda(\mathbf{x}_i))$$

We have $\mathbb{E}Y_i = \lambda(\mathbf{x}_i)$. The NN takes $\mathbf{x}_i$ & predicts $\mathbb{E}Y_i$.

Note

For insurance, *this is a bit weird*. The exposures are different for each policy.

$\lambda(\mathbf{x}_i)$ is the expected number of claims for the duration of policy i's contract.

Normally, $\text{Exposure}_i \notin \mathbf{x}_i$, and $\lambda(\mathbf{x}_i)$ is the expected rate *per year*, then

$$Y_i \sim \mathsf{Poisson}(\text{Exposure}_i \times \lambda(\mathbf{x}_i)).$$

What values do we see in the data?

```
1 X_train_raw["Area"].value_counts()
2 X_train_raw["VehBrand"].value_counts()
3 X_train_raw["VehGas"].value_counts()
4 X_train_raw["Region"].value_counts()
```

```
Area
C    5514
D    4116
     ...
B    2387
F     444
Name: count, Length: 6, dtype: int64
```

```
VehGas
Regular    10658
Diesel      8092
Name: count, dtype: int64
```

```
VehBrand
B1     4998
B2     4906
       ...
B11     283
B14     140
Name: count, Length: 11, dtype: int64
```

```
Region
R24    6493
R82    2112
       ...
R42      48
R43      26
Name: count, Length: 22, dtype: int64
```

How we preprocessed last time

```
ct = make_column_transformer(
  (OneHotEncoder(sparse_output=False, drop="first"), ["VehGas", "VehBrand", "Region"]),
  (OrdinalEncoder(), ["Area"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
```

```
X_train_raw.head(3)
```

	Exposure	Area	VehPower	VehAge
0	1.00	A	7.0	8.0
1	0.79	B	7.0	7.0
2	1.00	C	6.0	13.0

```
X_train.head(3)
```

	VehGas_Regular	VehBrand_B10	Ve
0	0.0	0.0	0.(
1	0.0	0.0	0.(
2	1.0	0.0	0.(

3 rows × 39 columns

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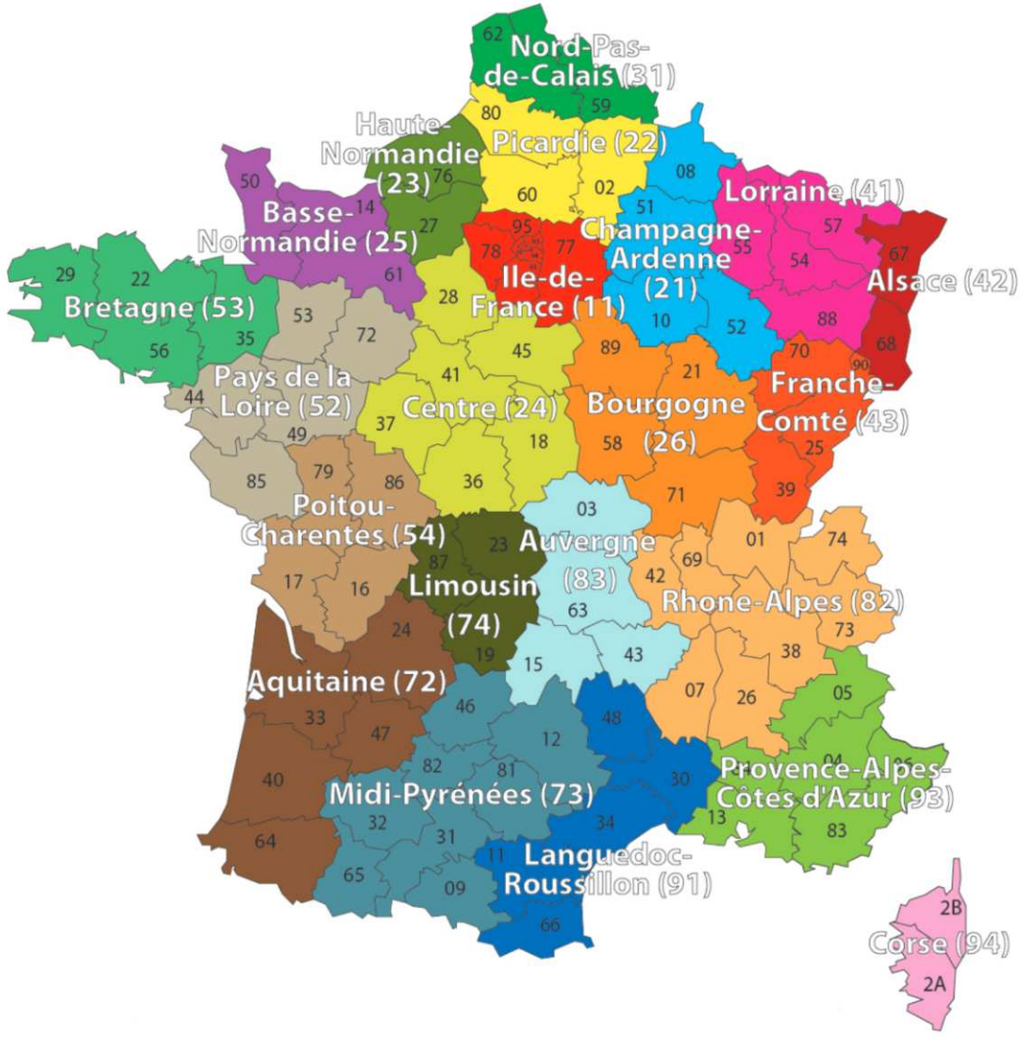
Region column



French Administrative Regions

Source: Wikimedia

One-hot encoding

```
1 oh = OneHotEncoder(sparse_output=False)
2 X_train_oh = oh.fit_transform(X_train_raw[["Region"]])
3 X_test_oh = oh.transform(X_test_raw[["Region"]])
4 print(list(X_train_raw["Region"][:5]))
5 X_train_oh.head()
```

```
['R24', 'R21', 'R53', 'R24', 'R82']
```

	Region_R11	Region_R21	Region_R22	Region_R23	Region_R24	Region_R2
0	0.0	0.0	0.0	0.0	1.0	0.0
1	0.0	1.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...
3	0.0	0.0	0.0	0.0	1.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0

5 rows × 22 columns

Train on one-hot inputs

```
num_regions = len(oh.categories_[0])

random.seed(12)
model = Sequential([
  Input(shape=(num_regions,)),
  Dense(2),
  Dense(1, activation="exponential")
])

model.compile(optimizer="adam", loss="poisson")

es = EarlyStopping(verbose=True)
hist = model.fit(X_train_oh, y_train, epochs=100, verbose=0, validation_split=0.2, callba

hist.history["val_loss"][-1]
```

```
Epoch 6: early stopping
```

```
0.7679625153541565
```

Make a fake batch of data

```
1 X = np.eye(num_regions)
2 pd.DataFrame(X, columns=oh.categories_[0]
```

	R11	R21	R22	R23	R24	R25	R2
0	1.0	0.0	0.0	0.0	0.0	0.0	0.(
1	0.0	1.0	0.0	0.0	0.0	0.0	0.(
...	...	...	...	...	...	...	...
20	0.0	0.0	0.0	0.0	0.0	0.0	0.(
21	0.0	0.0	0.0	0.0	0.0	0.0	0.(

22 rows × 22 columns

```
1 model.layers[0](X)
```

```
tensor([[-0.0646, -0.2346],
        [ 0.1463, -0.0323],
        [-0.3050, -0.2913],
        [ 0.0257, -0.3189],
        [ 0.3848, -0.2778],
        [ 0.0983, -0.4137],
        [ 0.2235, -0.2502],
        [ 0.1192,  0.3202],
        [ 0.6295, -0.4385],
        [ 0.2008, -0.0306],
        [-0.2030, -0.3797],
        [ 0.7612, -0.2256],
        [ 0.5068, -0.0436],
        [ 0.7867, -0.4326],
        [-0.2026, -0.1788],
        [-0.2430, -0.0941],
```

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The first layer

```
1 layer = model.layers[0]
2 W, b = layer.get_weights()
3 X.shape, W.shape, b.shape
```

```
((22, 22), (22, 2), (2,))
```

```
1 X @ W + b
```

```
array([[-0.06, -0.23],
       [ 0.15, -0.03],
       [-0.3 , -0.29],
       [ 0.03, -0.32],
       [ 0.38, -0.28],
       [ 0.1 , -0.41],
       [ 0.22, -0.25],
       [ 0.12,  0.32],
       [ 0.63, -0.44],
       [ 0.2 , -0.03],
       [-0.2 , -0.38],
       [ 0.76, -0.23],
       [ 0.51, -0.04],
       [ 0.79, -0.43],
       [-0.2 , -0.18],
       [-0.24, -0.09],
```

```
1 W + b
```

```
array([[-0.06, -0.23],
       [ 0.15, -0.03],
       [-0.3 , -0.29],
       [ 0.03, -0.32],
       [ 0.38, -0.28],
       [ 0.1 , -0.41],
       [ 0.22, -0.25],
       [ 0.12,  0.32],
       [ 0.63, -0.44],
       [ 0.2 , -0.03],
       [-0.2 , -0.38],
       [ 0.76, -0.23],
       [ 0.51, -0.04],
       [ 0.79, -0.43],
       [-0.2 , -0.18],
       [-0.24, -0.09],
```

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Just a look-up operation

```
1 display(list(oh.categories_[0]))
```

```
['R11',
 'R21',
 'R22',
 'R23',
 'R24',
 'R25',
 'R26',
 'R31',
 'R41',
 'R42',
 'R43',
 'R52',
 'R53',
 'R54',
 'R72',
 'R73',
```

```
1 W + b
```

```
array([[-0.06, -0.23],
       [ 0.15, -0.03],
       [-0.3 , -0.29],
       [ 0.03, -0.32],
       [ 0.38, -0.28],
       [ 0.1 , -0.41],
       [ 0.22, -0.25],
       [ 0.12,  0.32],
       [ 0.63, -0.44],
       [ 0.2 , -0.03],
       [-0.2 , -0.38],
       [ 0.76, -0.23],
       [ 0.51, -0.04],
       [ 0.79, -0.43],
       [-0.2 , -0.18],
       [-0.24, -0.09],
```

Each category maps to one row of `W + b` — that row *is* its embedding. No matrix multiply needed.

Turn the region into an index

```
oe = OrdinalEncoder()
X_train_reg = oe.fit_transform(X_train_raw[["Region"]])
X_test_reg = oe.transform(X_test_raw[["Region"]])

for i, reg in enumerate(oe.categories_[0][:3]):
  print(f"The Region value {reg} gets turned into {i}.")
```

```
The Region value R11 gets turned into 0.
The Region value R21 gets turned into 1.
The Region value R22 gets turned into 2.
```

Use an Embedding layer

```
num_regions = X_train_raw["Region"].nunique()

random.seed(12)
model = Sequential([
  Input(shape=(1,)),
  Embedding(input_dim=num_regions, output_dim=2),
  Dense(1, activation="exponential")
])

model.compile(optimizer="adam", loss="poisson")
```

```
es = EarlyStopping(verbose=True)
hist = model.fit(X_train_reg, y_train, epochs=100, verbose=0,
    validation_split=0.2, callbacks=[es])
hist.history["val_loss"][-1]
```

```
Epoch 4: early stopping
0.7677991390228271
```

```
model.layers
```

```
[<Embedding name=embedding, built=True>, <Dense name=dense_2, built=True>]
```

Aside: the output here is shaped `(None, 1, 1)`, not `(None, 1)` — a harmless extra axis from the `Embedding` that we will tidy up later.

Keras' Embedding Layer

```
model.layers[0].get_weights()[0]
```

```
array([[ 0.05, -0.07],
       [-0.02,  0.03],
       [-0.04, -0.02],
       [ 0.09, -0.12],
       [ 0.24, -0.22],
       [ 0.16, -0.19],
       [ 0.17, -0.17],
       [-0.05,  0.09],
       [ 0.36, -0.33],
       [ 0.03, -0.01],
       [ 0.02, -0.07],
       [ 0.36, -0.3 ],
       [ 0.21, -0.16],
       [ 0.42, -0.37],
       [-0.02, -0.02],
       [-0.06,  0.02],
```

```
X_train_raw["Region"].head(4)
```

```
0    R24
1    R21
2    R53
3    R24
Name: Region, dtype: str
```

```
X_sample = X_train_reg[:4].to_numpy()
X_sample
```

```
array([[ 4.],
       [ 1.],
       [12.],
       [ 4.]])
```

```
enc_tensor = model.layers[0](X_sample)
keras.ops.convert_to_numpy(enc_tensor).sq
```

```
array([[ 0.24, -0.22],
       [-0.02,  0.03],
       [ 0.21, -0.16],
       [ 0.24, -0.22]], dtype=float32)
```

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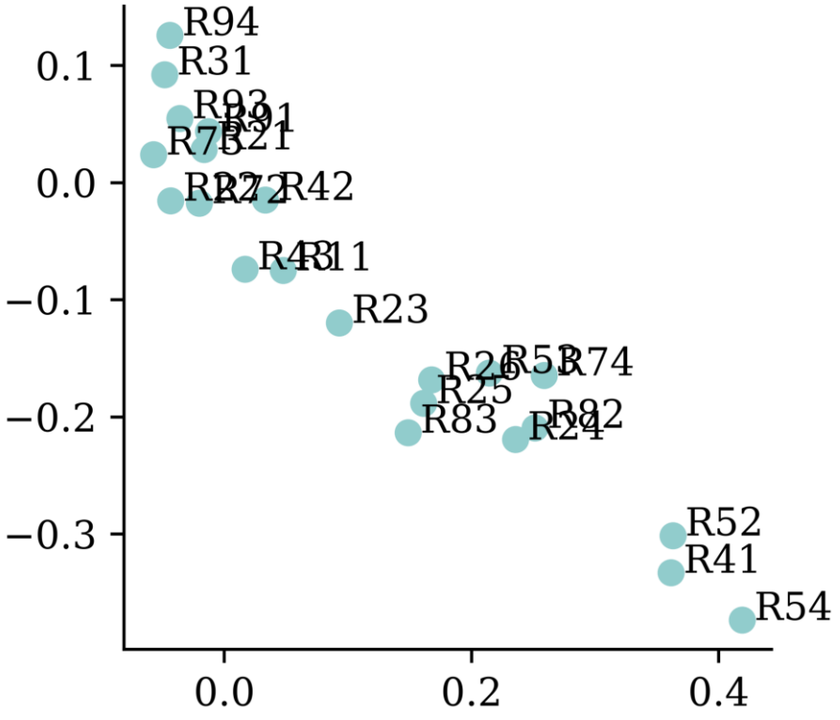
The embedding dimension

- Entity embedding is especially useful when there is a very large number of categories (such as 1,000 or 10,000) and you want to reduce the number of columns.
- The embedding dimension is really a *hyperparameter to tune* (e.g. by validation loss); there is no settled formula for it.
- One heuristic is to use $n^{1/4}$ for a variable with n categories (The TensorFlow Team, 2017), but there's no theoretical justification, and others disagree.
- In this case, a nice side benefit is that 2-dimensional embeddings can be plotted directly.

The learned embeddings

```
points = model.layers[0].get_weights()[0]
plt.figure(figsize=(3, 3))
plt.scatter(points[:,0], points[:,1])
for i in range(num_regions):
  plt.text(points[i,0]+0.01, points[i,1]
```



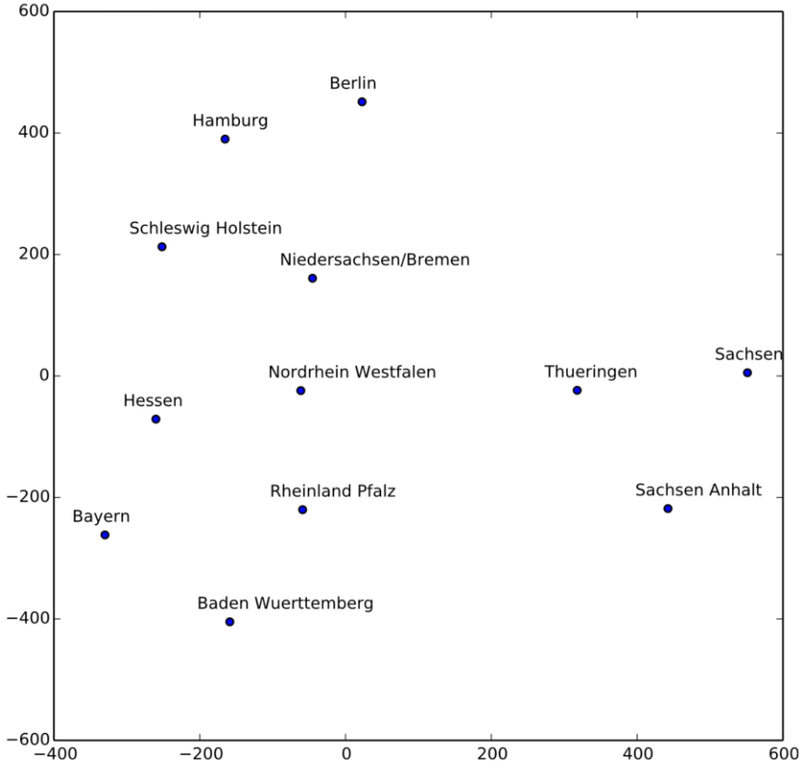
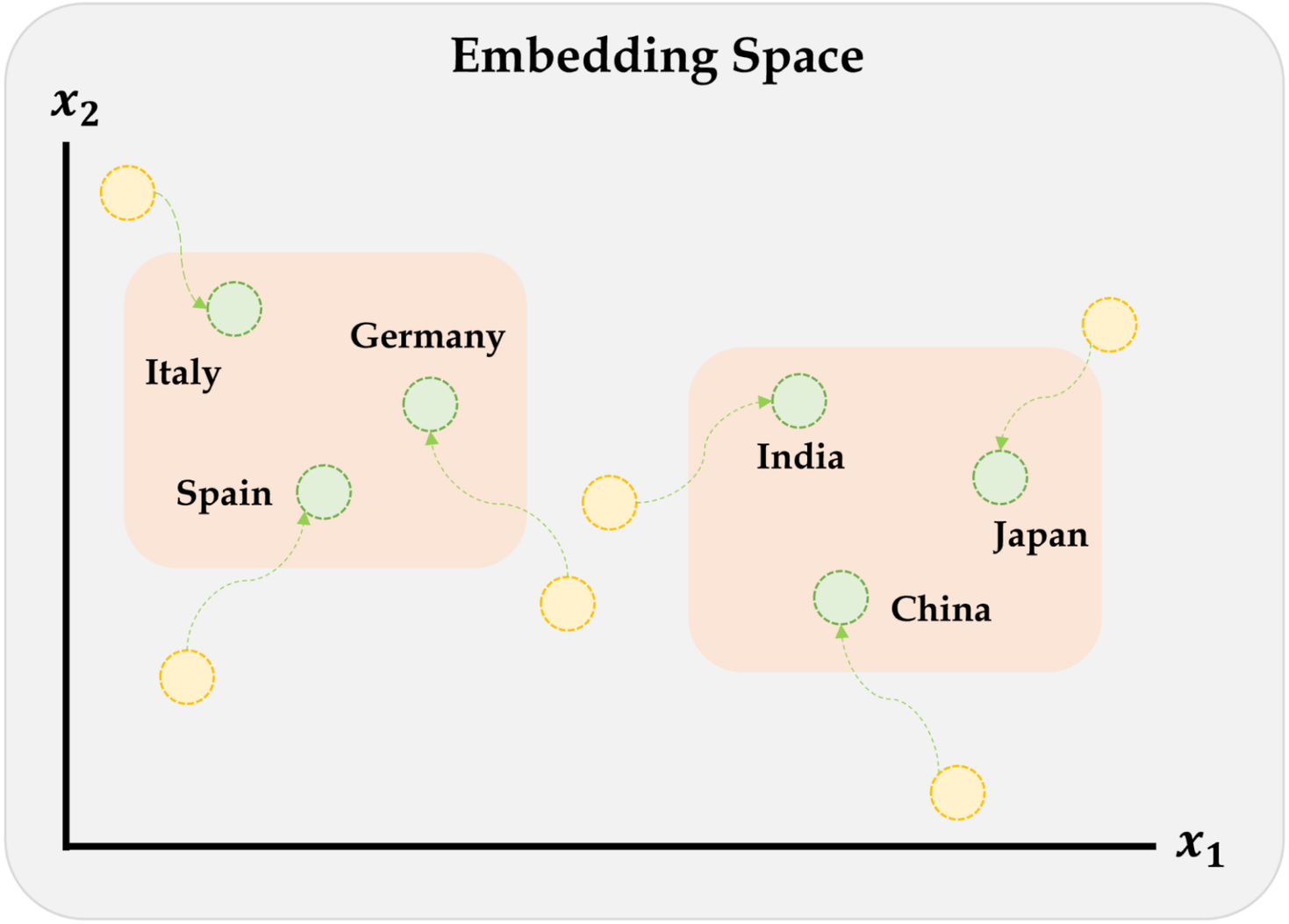


FIG. 3. The learned German state embedding is mapped to a 2D space with t-SNE. The relative positions of German states here resemble that on the real German map surprisingly well.

Reproduced from the Guo & Berkhahn (2016, p. 7).

Embeddings improve during training



Embeddings move during training to wherever they need to be to minimise training loss.

Source: Marcus Lautier (2022).

Embeddings & other inputs

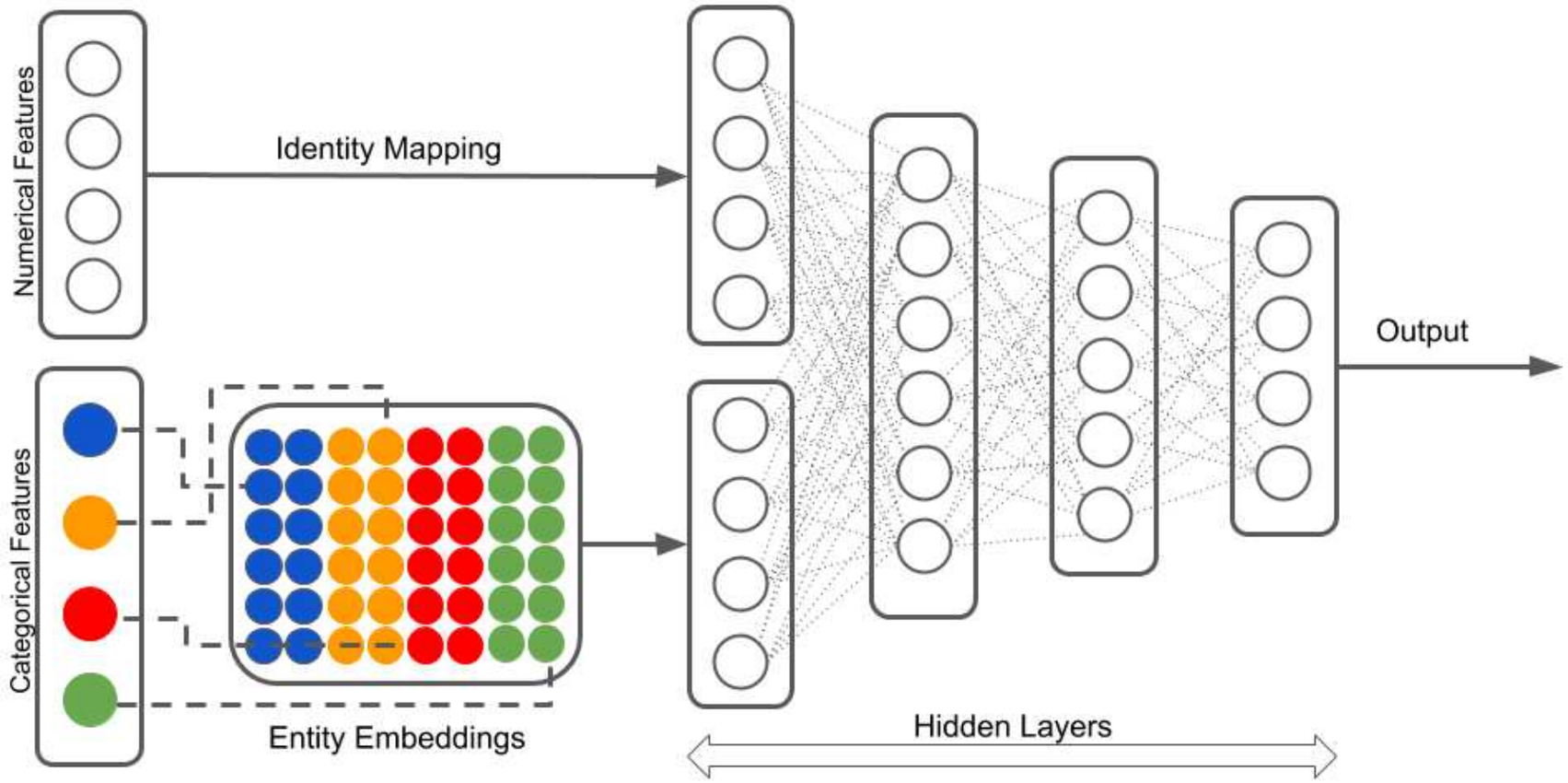


Illustration of a neural network with both continuous and categorical inputs.

We can't do this with Sequential models...

Source: LotusLabs Blog, Accurate insurance claims prediction with Deep Learning.

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Converting Sequential models

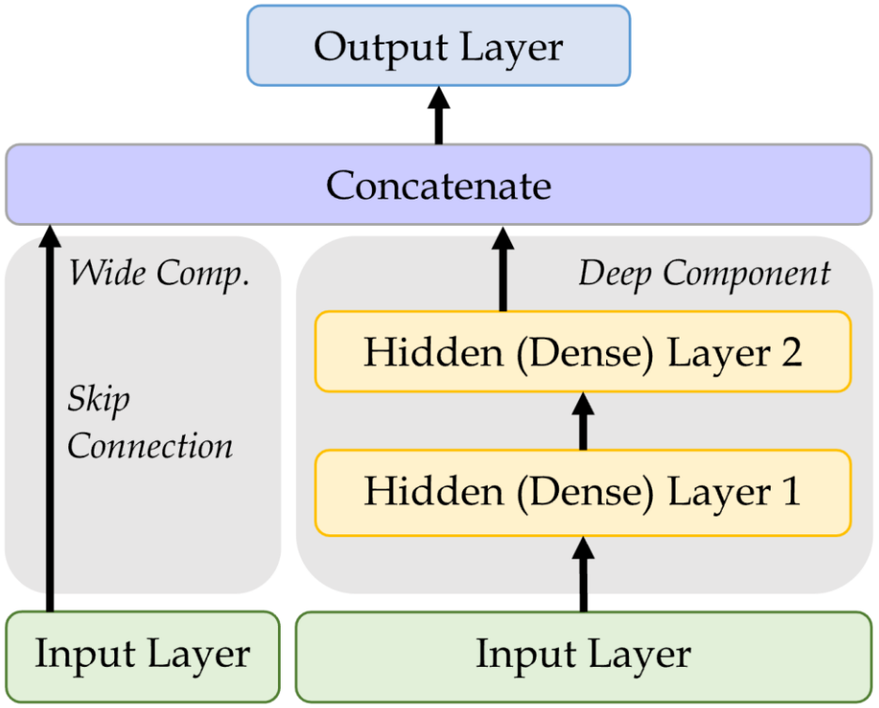
```
random.seed(12)

model = Sequential([
  Input(shape=(X_train_oh.shape[1],)),
  Dense(30, "leaky_relu"),
  Dense(1, "exponential")
])

model.compile(
  optimizer="adam",
  loss="poisson")

hist = model.fit(
  X_train_oh, y_train,
  epochs=1, verbose=0,
  validation_split=0.2)
hist.history["val_loss"][-1]
```

```
0.7695862650871277
```

```
random.seed(12)

inputs = Input(shape=(X_train_oh.shape[1]
x = Dense(30, "leaky_relu")(inputs)
out = Dense(1, "exponential")(x)
model = Model(inputs, out)

model.compile(
  optimizer="adam",
  loss="poisson")

hist = model.fit(
  X_train_oh, y_train,
  epochs=1, verbose=0,
  validation_split=0.2)
hist.history["val_loss"][-1]
```

```
0.7695212364196777
```

Wide & Deep network



An illustration of the wide & deep network architecture.

Add a *skip connection* from input to output layers.

```
inp = Input(shape=X_train.shape[1:])
hidden1 = Dense(30, "leaky_relu")(inp)
hidden2 = Dense(30, "leaky_relu")(hidden1)
concat = Concatenate()(
  [inp, hidden2])
output = Dense(1)(concat)
model = Model(inputs=[inp], outputs=[output])
```

Sources: Marcus Lautier (2022) & Aurélien Géron (2019), *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 2nd Edition, Chapter 10 code snippet

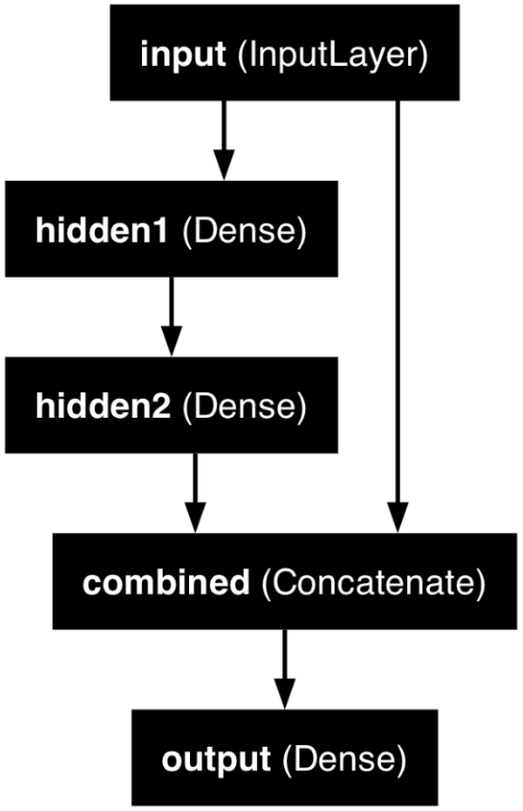
Naming the layers

For complex networks, it is often useful to give meaningful names to the layers.

```
input_ = Input(shape=X_train.shape[1:], name="input")
hidden1 = Dense(30, activation="leaky_relu", name="hidden1")(input_)
hidden2 = Dense(30, activation="leaky_relu", name="hidden2")(hidden1)
concat = Concatenate(name="combined")([input_, hidden2])
output = Dense(1, name="output")(concat)
model = Model(inputs=[input_], outputs=[output])
```

Inspecting a complex model

```
1 plot_model(model, show
```




```
1 model.summary(line_length=60)
```

Model: "functional_5"

Layer (type)	Output Shape	Param #	Connected to
input (InputLayer)	(None, 39)	0	–
hidden1 (Dense)	(None, 30)	1,200	input[0][0]
hidden2 (Dense)	(None, 30)	930	hidden1[0][0]
combined (Concatenate)	(None, 69)	0	input[0][0], hidden2[0][0]
output (Dense)	(None, 1)	70	combined[0][…

Total params: 2,200 (8.59 KB)
Trainable params: 2,200 (8.59 KB)
Non-trainable params: 0 (0.00 B)

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The desired architecture

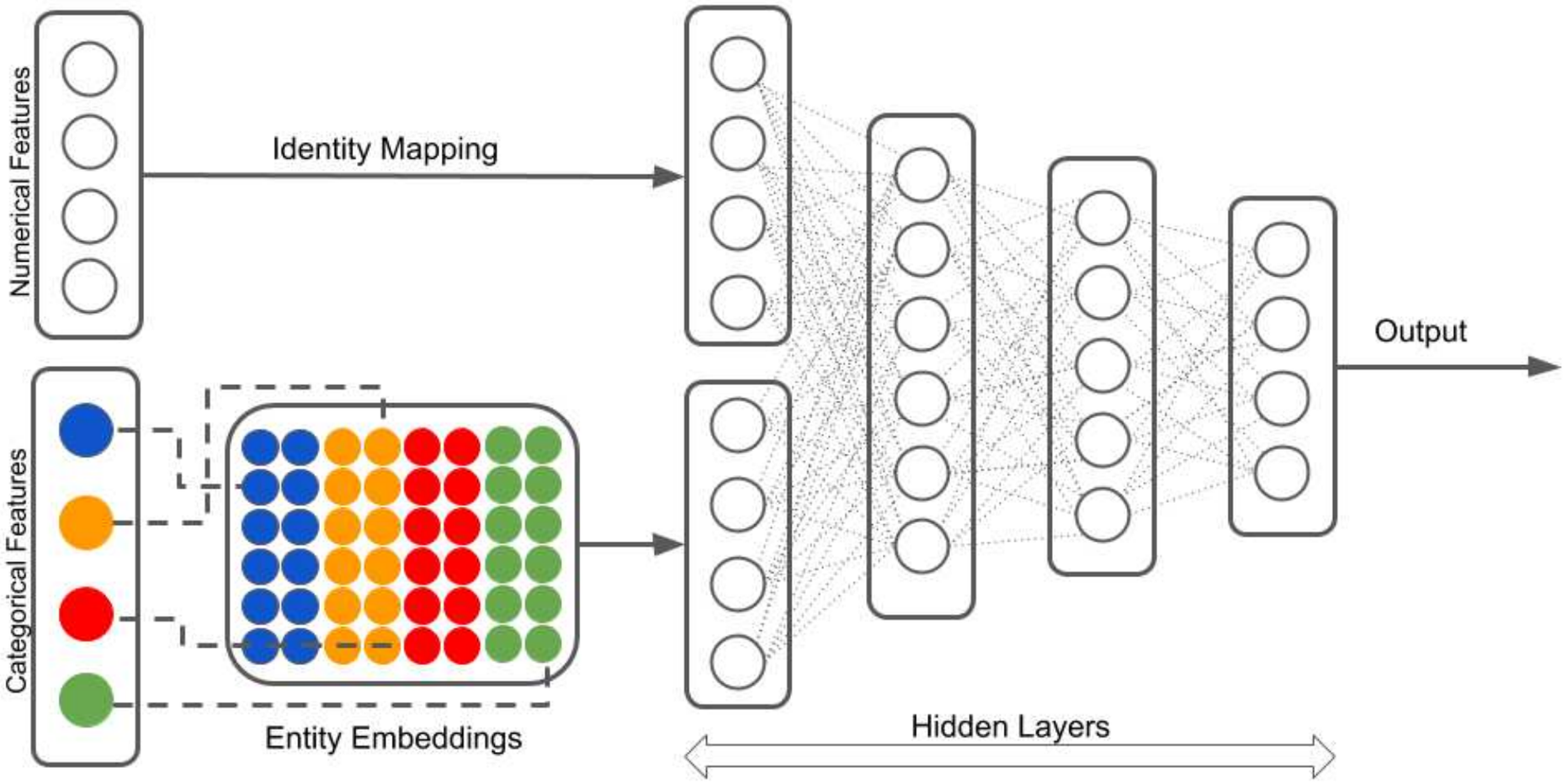


Illustration of a neural network with both continuous and categorical inputs.

Source: LotusLabs Blog, Accurate insurance claims prediction with Deep Learning.

Preprocess all French motor inputs

Transform the categorical variables to integers:

```
num_brands, num_regions = X_train_raw[["VehBrand", "Region"]].nunique()

ct = make_column_transformer(
  (OrdinalEncoder(), ["VehBrand", "Region", "Area", "VehGas"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
X_test = ct.transform(X_test_raw)
```

Split the brand and region data apart from the rest:

```
X_train_brand = X_train["VehBrand"]
X_train_region = X_train["Region"]
X_train_rest = X_train.drop(["VehBrand", "Region"], axis=1)

X_test_brand = X_test["VehBrand"]
X_test_region = X_test["Region"]
X_test_rest = X_test.drop(["VehBrand", "Region"], axis=1)
```

Organise the inputs

Make a Keras `Input` for: vehicle brand, region, & others.

```
veh_brand = Input(shape=(1,), name="veh_brand")
region = Input(shape=(1,), name="region")
other_inputs = Input(shape=X_train_rest.shape[1:], name="other_inputs")
```

Create embeddings and join them with the other inputs.

```
random.seed(1337)
veh_brand_ee = Embedding(input_dim=num_brands, output_dim=2,
    name="veh_brand_ee")(veh_brand)
veh_brand_ee = Reshape(target_shape=(2,))(veh_brand_ee)

region_ee = Embedding(input_dim=num_regions, output_dim=2, name="region_ee")(region)
region_ee = Reshape(target_shape=(2,))(region_ee)

x = Concatenate(name="combined")([veh_brand_ee, region_ee, other_inputs])
```

Complete the model and fit it

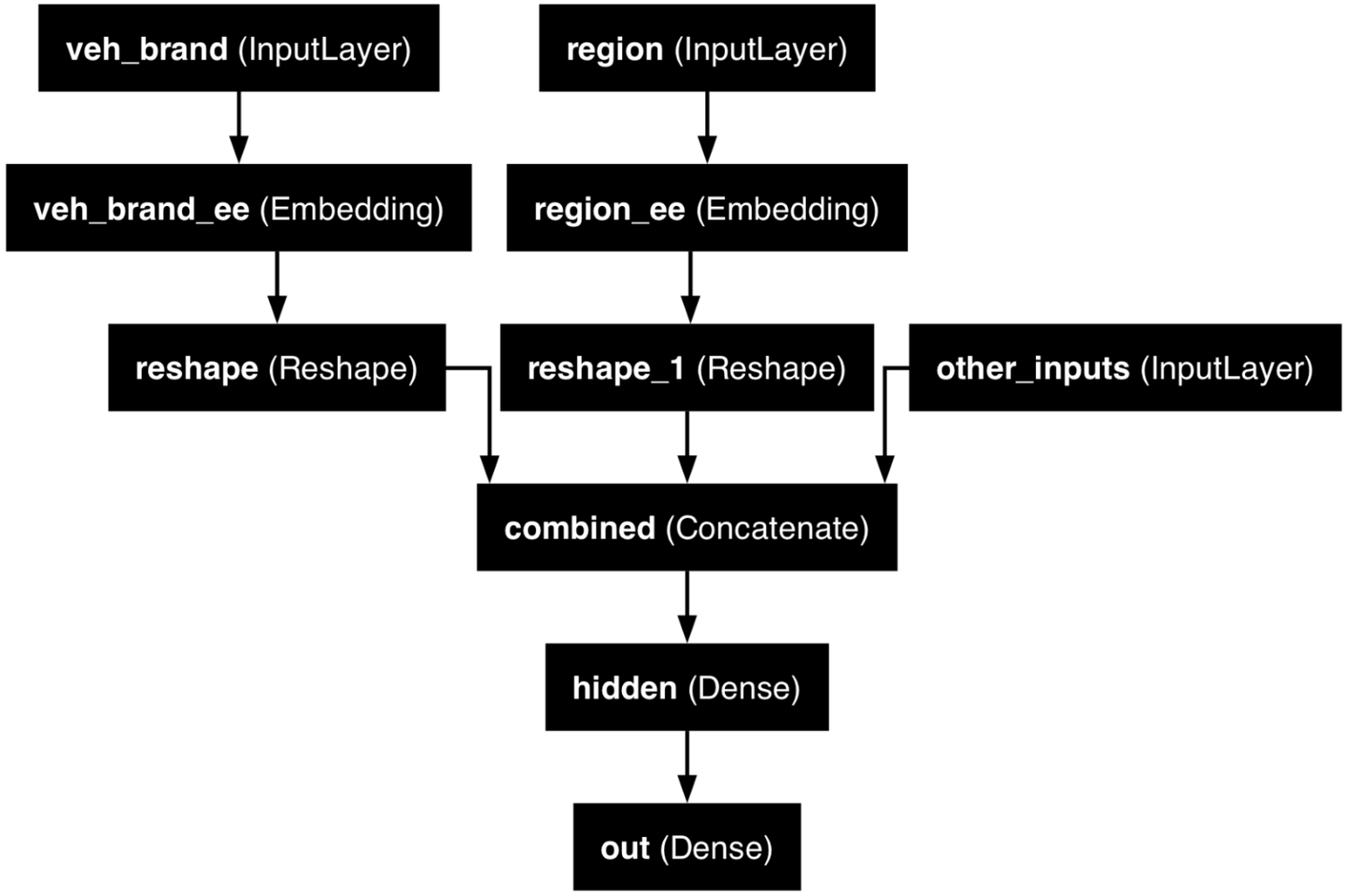
Feed the combined embeddings & continuous inputs to some normal dense layers.

```
x = Dense(30, "relu", name="hidden")(x)
out = Dense(1, "exponential", name="out")(x)

model = Model([veh_brand, region, other_inputs], out)
model.compile(optimizer="adam", loss="poisson")

hist = model.fit((X_train_brand, X_train_region, X_train_rest),
    y_train, epochs=100, verbose=0,
    callbacks=[EarlyStopping(patience=5)], validation_split=0.2)
np.min(hist.history["val_loss"])
```

```
np.float64(0.6855398416519165)
```

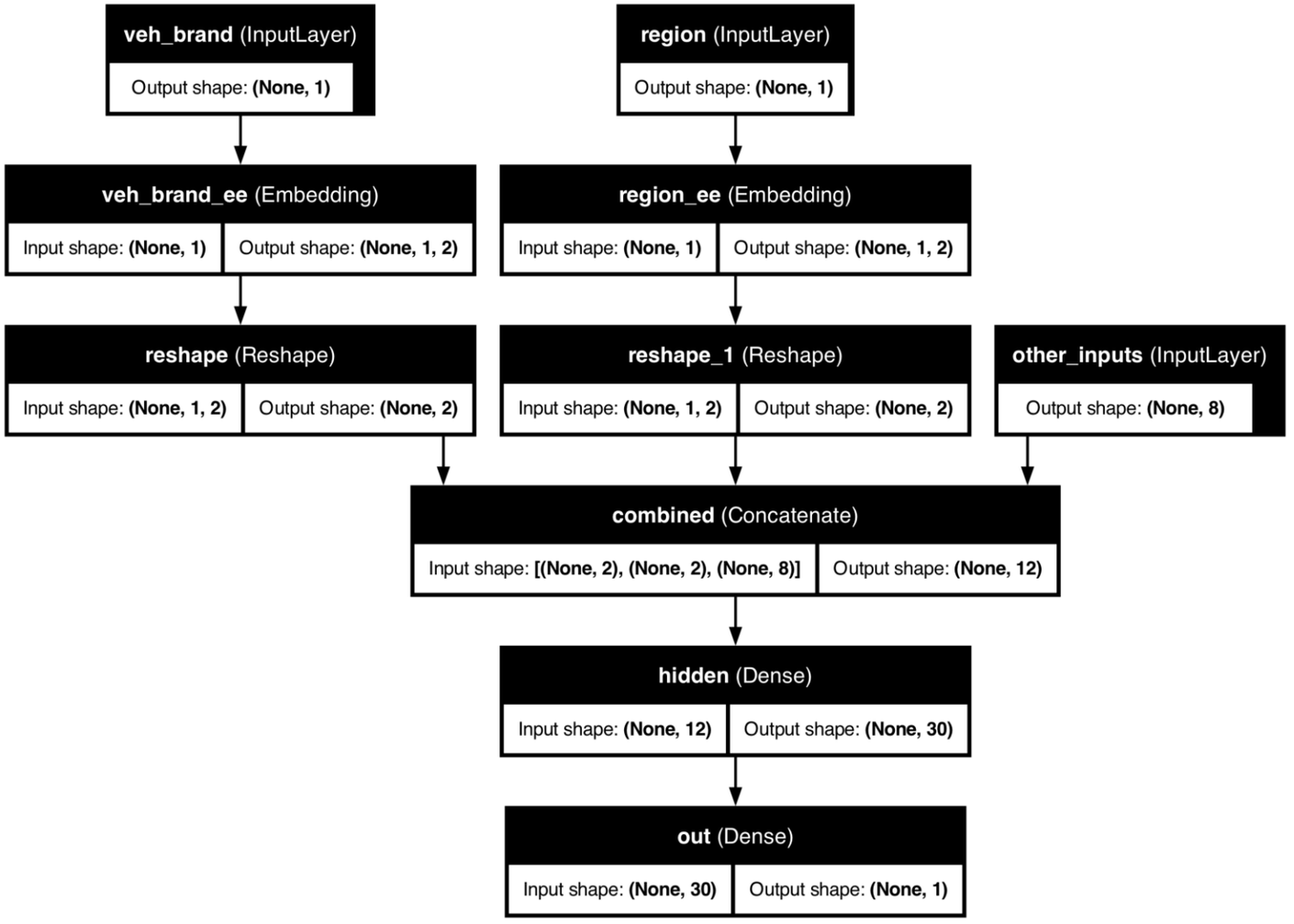
Plotting this model

```
1 plot_model(model, show_layer_names=True)
```



Why we need to reshape

```
1 plot_model(model, show_layer_names=True, show_shapes=True)
```



`Embedding` turns `(None, 1)` into `(None, 1, 2)`; `Reshape` drops the extra axis to `(None, 2)` so we can `Concatenate`.

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Lecture Outline

- Entity Embedding
- The French Motor Dataset
- From One-Hot Encoding to Embeddings
- Keras' Functional API
- French Motor Dataset with Embeddings
- **Scale by Exposure**

Two different models

Have $\{(\mathbf{x}_i, y_i)\}_{i=1,\ldots,n}$ for $\mathbf{x}_i \in \mathbb{R}^{47}$ and $y_i \in \mathbb{N}_0$.

Model 1: Say $Y_i \sim \mathsf{Poisson}(\lambda(\mathbf{x}_i))$.

But, the exposures are different for each policy. $\lambda(\mathbf{x}_i)$ is the expected number of claims for the duration of policy i's contract.

Model 2: Say $Y_i \sim \mathsf{Poisson}(\text{Exposure}_i \times \lambda(\mathbf{x}_i))$.

Now, $\text{Exposure}_i \notin \mathbf{x}_i$, and $\lambda(\mathbf{x}_i)$ is the rate *per year*.

Just take continuous variables

```
ct = make_column_transformer(
  ("passthrough", ["Exposure"]),
  ("drop", ["VehBrand", "Region", "Area", "VehGas"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
X_test = ct.transform(X_test_raw)
```

Split exposure apart from the rest:

```
X_train_exp = X_train["Exposure"]
X_test_exp = X_test["Exposure"]
X_train_rest = X_train.drop("Exposure", axis=1)
X_test_rest = X_test.drop("Exposure", axis=1)
```

Organise the inputs:

```
exposure = Input(shape=(1,), name="exposure")
other_inputs = Input(shape=X_train_rest.shape[1:], name="other_inputs")
```

Make & fit the model

Feed the continuous inputs to some normal dense layers.

```
random.seed(1337)
x = Dense(30, "relu", name="hidden1")(other_inputs)
x = Dense(30, "relu", name="hidden2")(x)
lambda_ = Dense(1, "exponential", name="lambda")(x)
```

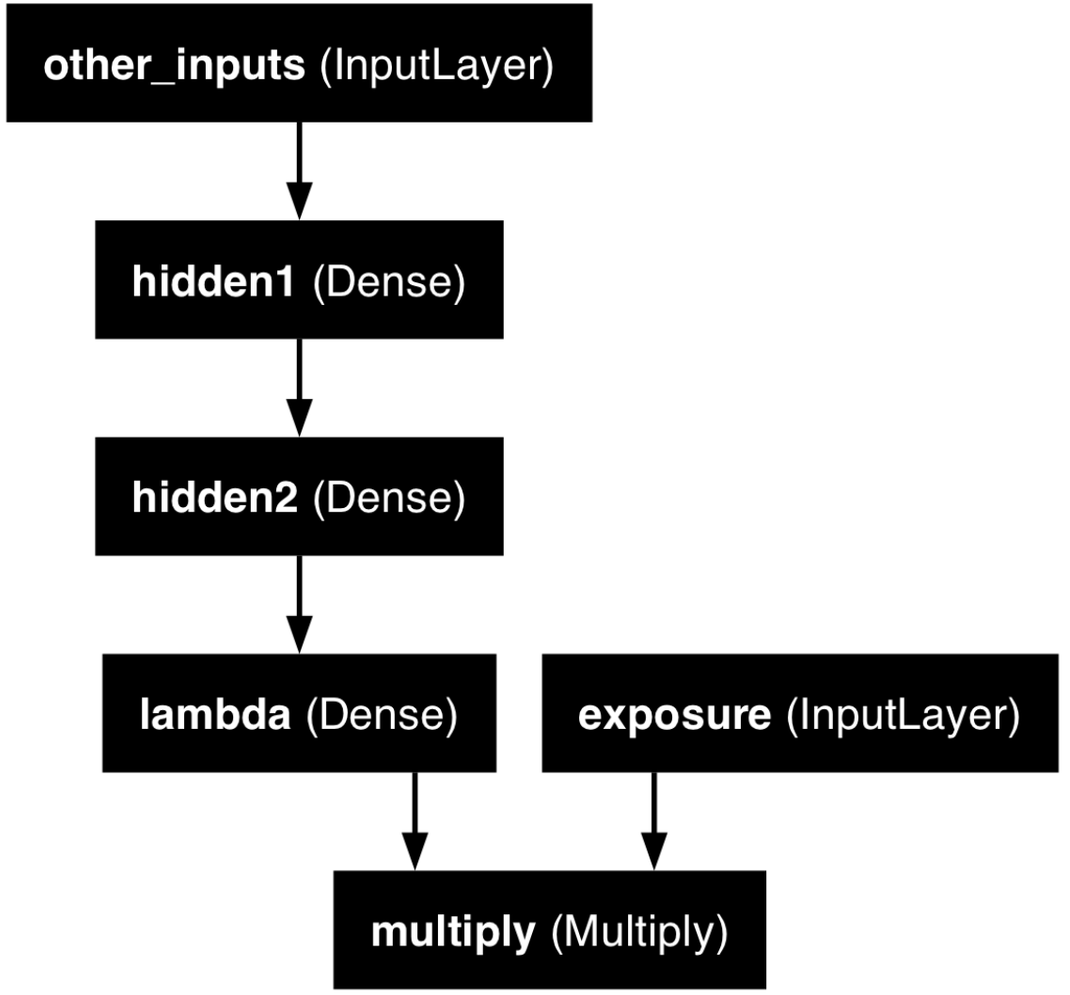
```
out = lambda_ * exposure
model = Model([exposure, other_inputs], out)
model.compile(optimizer="adam", loss="poisson")

es = EarlyStopping(patience=10, restore_best_weights=True, verbose=1)
hist = model.fit((X_train_exp, X_train_rest), y_train, epochs=100, verbose=0,
    callbacks=[es], validation_split=0.2)
np.min(hist.history["val_loss"])
```

```
Epoch 29: early stopping
Restoring model weights from the end of the best epoch: 19.
```

```
np.float64(0.9100556373596191)
```

Plot the model

```
1 plot_model(model, show_layer_names=True)
```



Further reading

Avanzi et al. (2024) focus on how to handle the case of high-cardinality categorical features, and propose a new architecture specifically tailored for actuarial purposes.

Package Versions

```
1 from watermark import watermark
2 print(watermark(python=True, packages="keras,matplotlib,numpy,pandas,seaborn,scipy,torch"
```

```
Python implementation: CPython
Python version       : 3.14.5
IPython version      : 9.15.0

keras     : 3.15.0
matplotlib: 3.11.0
numpy     : 2.5.0
pandas    : 3.0.3
seaborn   : 0.13.2
scipy     : 1.18.0
torch     : 2.12.1
```

Glossary

- entity embeddings
- Input layer
- Keras functional API
- Reshape layer
- skip connection
- wide & deep network

References

Avanzi, B., Taylor, G., Wang, M., & Wong, B. (2024). Machine learning with high-cardinality categorical features in actuarial applications. *ASTIN Bulletin: The Journal of the IAA*, *54*(2), 213–238. https://doi.org/10.1017/asb.2024.7

Guo, C., & Berkhahn, F. (2016). Entity embeddings of categorical variables. *arXiv Preprint arXiv:1604.06737*.

Noll, A., Salzmann, R., & Wüthrich, M. V. (2020). Case study: French motor third-party liability claims. *SSRN Manuscript 3164764*.

The TensorFlow Team. (2017). *Introducing TensorFlow feature columns*. Google Developers Blog. https://developers.googleblog.com/introducing-tensorflow-feature-columns/